

Spherical Mirrors

Curve a mirror inward or outward and the image changes — that's a concave or convex mirror at work.

Concave and convex mirrors

Mirrors with a **spherical (curved) reflecting surface** are **spherical mirrors** — like the two sides of a shiny spoon.

Concave mirror	Convex mirror
Reflecting surface curves inward	Reflecting surface curves outward
Close object → enlarged, erect ; far object → inverted	Always erect & diminished (smaller)

Where they are used

- **Concave:** torch/headlight reflectors, dentist's mirror (enlarged view of teeth), reflecting telescopes, solar concentrators.
- **Convex:** vehicle side-view mirrors, road-safety mirrors at bends, shop surveillance mirrors — they give an erect image and a **wide view**.

"Objects are closer than they appear": a convex side mirror makes vehicles look small and far, so the warning reminds you they're actually nearer.

Compare with a plane mirror: a plane mirror always gives an erect image of the *same* size; spherical mirrors change the size (and concave can invert it).

Where students lose marks

Trap 1 — Concave curves IN, convex curves OUT. Concave = "caved in".

Trap 2 — Convex always gives erect, diminished images. Concave can give enlarged or inverted images depending on distance.

Trap 3 — Match mirror to use. Wide view/safety → convex; concentrate light or enlarge → concave.

Model answers — write them like this

Example A. Why are convex mirrors used as side-view mirrors in vehicles?

Step 1: A convex mirror always forms an erect, diminished image.

Step 2: Because it curves outward, it covers a wider area behind.

∴ They give an upright, wide view of traffic behind.

Example B. Which mirror does a dentist use to see an enlarged image of a tooth, and why?

Step 1: An enlarged, erect image of a close object is needed.

Step 2: A concave mirror gives this when the object is close.

∴ A concave mirror, held close to the tooth.

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